

Database Design

Document Information

Project: Blogging System API

Version: 1.0

Introduction

This document describes the database design for the Blogging System API.

The application uses **PostgreSQL** as its relational database management system and **Entity Framework Core** as its Object-Relational Mapper (ORM).

The database has been normalized to reduce redundancy while maintaining efficient relationships between entities.

Design Principles

The database is designed according to the following principles:

- Third Normal Form (3NF)
 - Referential Integrity
 - Primary and Foreign Key Constraints
 - UUID (GUID) Primary Keys
 - Consistent Naming Conventions
 - Timestamp Tracking
 - Secure Authentication Storage
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Database Overview

Version 1 of the application consists of the following tables:

Table	Purpose
Users	Stores registered users

Table	Purpose
Posts	Stores blog posts
Comments	Stores comments and replies
RefreshTokens	Stores JWT refresh tokens
Otps	Stores email verification and password reset OTPs

Users Table

Description

Stores all registered users.

Columns

Column	Type	Constraints
Id	UUID	Primary Key
FirstName	VARCHAR(100)	NOT NULL
Surname	VARCHAR(100)	NOT NULL
Email	VARCHAR(255)	UNIQUE, NOT NULL
PasswordHash	TEXT	NOT NULL
Avatar	TEXT	NULL
Bio	TEXT	NULL
Role	ENUM	NOT NULL
IsVerified	BOOLEAN	NOT NULL
CreatedAt	TIMESTAMP WITH TIME ZONE	NOT NULL
UpdatedAt	TIMESTAMP WITH TIME ZONE	NOT NULL

Posts Table

Description

Stores blog posts created by administrators.

Columns

Column	Type	Constraints
Id	UUID	Primary Key
Title	VARCHAR(250)	NOT NULL
Summary	TEXT	NOT NULL
Content	TEXT	NOT NULL
Status	ENUM	NOT NULL
CommentsLocked	BOOLEAN	NOT NULL
AuthorId	UUID	Foreign Key → Users(Id)
CreatedAt	TIMESTAMP WITH TIME ZONE	NOT NULL
UpdatedAt	TIMESTAMP WITH TIME ZONE	NOT NULL

Comments Table

Description

Stores comments and threaded replies.

Columns

Column	Type	Constraints
Id	UUID	Primary Key
Content	TEXT	NOT NULL
UserId	UUID	Foreign Key → Users(Id)
PostId	UUID	Foreign Key → Posts(Id)
ParentCommentId	UUID	Nullable, Foreign Key → Comments(Id)
CreatedAt	TIMESTAMP WITH TIME ZONE	NOT NULL
UpdatedAt	TIMESTAMP WITH TIME ZONE	NOT NULL

RefreshTokens Table

Description

Stores refresh tokens issued during authentication.

Columns

Column	Type	Constraints
Id	UUID	Primary Key
UserId	UUID	Foreign Key → Users(Id)
Token	TEXT	NOT NULL
ExpiresAt	TIMESTAMP WITH TIME ZONE	NOT NULL
RevokedAt	TIMESTAMP WITH TIME ZONE	NULL
CreatedAt	TIMESTAMP WITH TIME ZONE	NOT NULL

Otps Table

Description

Stores one-time passwords used for account verification and password reset.

Columns

Column	Type	Constraints
Id	UUID	Primary Key
UserId	UUID	Foreign Key → Users(Id)
Code	VARCHAR(6)	NOT NULL
Purpose	ENUM	NOT NULL
IsUsed	BOOLEAN	NOT NULL
ExpiresAt	TIMESTAMP WITH TIME ZONE	NOT NULL
CreatedAt	TIMESTAMP WITH TIME ZONE	NOT NULL

Enumerations

UserRole

Administrator
User

PostStatus

Draft
Published

OtpPurpose

EmailVerification
PasswordReset

Relationships

Users → Posts

- One User can author many Posts.
- Every Post belongs to exactly one User.

Relationship:

User (1) ----- (*) Posts

Users → Comments

- One User can create many Comments.
- Every Comment belongs to one User.

User (1) ----- (*) Comments

Posts → Comments

- One Post can contain many Comments.
- Every Comment belongs to one Post.

Post (1) ----- (*) Comments

Comments → Comments

A Comment may reference another Comment as its parent.

This enables unlimited nesting.

Comment
|
└── ParentCommentId

Users → RefreshTokens

One User may own multiple Refresh Tokens.

User (1) ----- (*) RefreshTokens

Users → Otps

One User may receive multiple OTPs over time.

User (1) ----- (*) Otps

Cascade Delete Rules

Parent	Child	Rule
User	Posts	Cascade Delete
User	Comments	Cascade Delete
User	RefreshTokens	Cascade Delete
User	Otps	Cascade Delete
Post	Comments	Cascade Delete
Comment	Replies	Cascade Delete

Deleting a parent entity shall automatically remove its dependent child records.

Indexing Strategy

The following indexes shall be created to improve query performance:

Users

- Email (Unique)
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Posts

- AuthorId
 - Status
 - CreatedAt
-

Comments

- PostId
 - UserId
 - ParentCommentId
-

RefreshTokens

- UserId

- Token
-

Otps

- UserId
 - Code
 - Purpose
-

Data Integrity Rules

- Every email address must be unique.
 - Every Post must have an existing author.
 - Every Comment must belong to an existing Post.
 - Every Comment must belong to an existing User.
 - Every Reply must reference an existing parent Comment.
 - OTPs must expire after their configured lifetime.
 - Refresh Tokens must be revoked before they can no longer be reused.
 - Password hashes shall never be stored in plain text.
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Naming Conventions

The database shall use the following conventions:

- PascalCase for table names.
 - PascalCase for column names.
 - Singular table names.
 - UUID primary keys.
 - Foreign keys named using the `<EntityName>Id` convention.
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Database Summary

Table	Primary Key
Users	Id (UUID)
Posts	Id (UUID)
Comments	Id (UUID)
RefreshTokens	Id (UUID)

Table	Primary Key
Otps	Id (UUID)

The schema is normalized, extensible, and designed to support the current project requirements while allowing future enhancements without significant structural changes.